

Efficient and Flexible Low-Power NTT for Lattice-Based Cryptography

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Abstract—We present a complete hardware implementation of the low-power NTT architecture of Fritzmann and Sepulveda [1], targeting FPGA prototyping and ASIC synthesis/place-and-route using Cadence Genus/Innovus on UMC 130 nm. The RTL implements Montgomery and Barrett modular reduction, optimised Cooley-Tukey NTT and inverse NTT with integrated post-processing, clock gating, and operand isolation. The design supports $n \in \{256, 512, 1024\}$ and $q \in \{7681, 12289\}$. FPGA results show ≤ 142 mW total power (1221 LUTs, 25 DSPs) on the Zynq-7000 — the same device family used in [1]. ASIC achieves 25 MHz timing closure (WNS = +6 ps, SS corner) with 17.65 mW post-route power and zero DRC violations on UMC 130 nm.

Index Terms—NTT, Post-Quantum Cryptography, CRYSTALS-Kyber, NewHope, FPGA, ASIC, Montgomery Reduction, Barrett Reduction, Clock Gating, Operand Isolation

I. INTRODUCTION & BACKGROUND

Quantum computers threaten RSA and ECC; NIST has standardised lattice-based post-quantum schemes such as CRYSTALS-Kyber and CRYSTALS-Dilithium. Their core operation is polynomial multiplication in $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n + 1)$, accelerated from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \log n)$ via the Number Theoretic Transform (NTT): $c = \text{NTT}^{-1}(\text{NTT}(a) \circ \text{NTT}(b))$.

Fritzmann and Sepulveda [1] proposed the first low-power NTT ASIC supporting multiple parameter sets using three techniques: (i) rescheduling operations into stall cycles for single-port RAM; (ii) integrated Barrett (Algorithm 2) and Montgomery (Algorithm 3) modular reduction; (iii) post-processing folded into the last inverse-NTT round (Algorithm 5), saving at least $2n$ extra cycles. Power is reduced via clock gating on butterfly registers and operand isolation on multiplier inputs, achieving $> 30\%$ dynamic power reduction [1].

This work re-implements all algorithms of [1] in synthesisable Verilog-2005 and performs a full RTL-to-GDS flow on UMC 130 nm, enabling a quantitative comparison against the reference UMC 65 nm implementation.

II. ARCHITECTURE & RTL DESIGN

The NTT core is a memory-based Cooley-Tukey butterfly with a 12-state FSM, single-port RAM ($n/2 \times 32$ bits, two 16-bit coefficients per word), a twiddle-factor LUT ($\log_2 n + 1$ entries), five Montgomery and two Barrett reduction instances. Power techniques follow [1]: clock gates are inferred by Genus from `if (bf_act)` RTL enables; inverse-NTT alpha-multiplier inputs are held at zero during the forward NTT (operand isolation). All parameters (n , q , q' , LUT contents, scaling factors) are run-time configurable, supporting the three parameter sets in Table I without re-synthesis.

TABLE I
SUPPORTED NTT PARAMETER SETS

Scheme	n	q	ω_n
CRYSTALS-Kyber	256	7681	5685
NewHope-512	512	12289	3400
NewHope-1024	1024	12289	10302

Correctness was verified via a cycle-accurate Python reference model and Vivado 2025.1 behavioural simulation (forward NTT + NTT⁻¹ round-trip) across all three parameter sets.

III. FPGA RESULTS (ZYNQ-7000 xc7z020clg484-1)

The design was synthesised on the Xilinx Zynq-7020 — the same device family (Zynq-7000 / Zedboard) used in [1] — using Vivado 2025.2. Table II shows post-synthesis utilisation. The resource count is n -independent because a single butterfly unit iterates over all $n \log_2 n$ operations sequentially.

TABLE II
FPGA RESOURCE UTILISATION — ZYNQ-7000 xc7z020clg484-1 (VIVADO 2025.2)

Resource	Used	Avail.	Util. (%)	Paper [1]
LUT as Logic	953	53,200	1.79	—
LUT as Dist. RAM	268	17,400	1.54	—
LUT (total)	1,221	53,200	2.30	980
Registers (FF)	573	106,400	0.54	395
DSP48E1	25	220	11.36	26
CARRY4	81	13,300	0.61	—

TABLE III
FPGA POWER COMPARISON (ZYNQ-7020, 20 MHz, TYPICAL, SAIF-ANNOTATED) — OPTIMIZED

n	Op.	$P_{\text{static}}(\text{mW})$	$P_{\text{dyn}}(\text{mW})$	$P_{\text{total}}(\text{mW})$
256	NTT	131	10	141
	NTT ⁻¹	131	11	142
512	NTT	131	10	141
	NTT ⁻¹	131	11	142
1024	NTT	131	11	141
	NTT ⁻¹	131	11	142

Dynamic power is n -independent (10–11 mW), consistent with the single-butterfly architecture. NTT⁻¹ draws 1 mW more than NTT due to alpha-scaling multiplier activity in the last round — a trend also reported in [1].

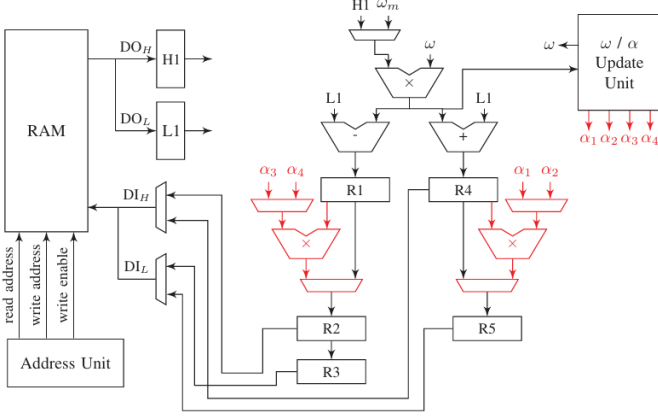


Fig. 1. Architecture implementation of paper [1]

IV. ASIC RESULTS (UMC 130 NM)

A. Synthesis (Cadence Genus)

Synthesis targeted 25 MHz (40 ns) at the TT corner (1.2 V, 25 °C). Because no SRAM macro was available in the PDK, the 512×32 -bit RAM was implemented as 33,500 flip-flops, which dominates area. Logic-only area is $178,473 \mu\text{m}^2$. Post-synthesis WNS was +472 ps.

B. Place-and-Route (Cadence Innovus)

The die ($1700 \times 1700 \mu\text{m}$, 52% utilisation) used 5 metal layers. Table IV shows timing closure progression; the pre-CTS violation of -13.1 ns is expected with ideal clocks. After CTS and post-CTS optimisation, timing is met at the SS worst-case corner (1.08 V, 125 °C) by 6 ps. The critical path traverses FSM $\rightarrow$ Montgomery multiply $\rightarrow$ Barrett subtract $\rightarrow$ alpha multiplier $\rightarrow$ RAM write, matching the double Montgomery-multiply critical path of [1].

TABLE IV
TIMING CLOSURE AT 25 MHz

Stage	WNS (ns)	Corner	Status
Post-synthesis	+0.472	TT, 1.2 V, 25 °C	MET
Pre-CTS	-13.117	SS, 1.08 V, 125 °C	Expected
Post-CTS	+0.006	SS, 1.08 V, 125 °C	MET

Post-route power at the SS corner is given in Table V. Sequential logic (47%) dominates due to the FF-based RAM. Zero DRC violations remain after ecoRoute passes.

TABLE V
POST-ROUTE POWER (UMC 130 NM, SS, 1.08 V, 125 °C)

Component	Power (mW)	Share (%)
Sequential (registers)	8.312	47.1
Combinational (logic)	7.434	42.1
Clock tree	1.908	10.8
Internal Switching	10.802	61.2
Leakage	6.054	34.3
Total	17.653	100.0

TABLE VI
COMPARISON WITH FRITZMANN & SEPULVEDA [1]

Metric	[1]	This Work
FPGA device	Zynq-7000 (Zedboard)	Zynq-7020
FPGA LUTs	980	1,221
FPGA DSPs	26	25
ASIC technology	UMC 65 nm	UMC 130 nm
ASIC frequency	25 MHz	25 MHz
ASIC total cells	14,352	45,120
ASIC area (μm^2)	329,598	854,844
Logic-only (μm^2)	$\approx 300\text{k}$	178,473
RAM type	SRAM macro	FF-based
P_{static}	19.2 μW	0.797 mW
P_{dynamic}	2.920 mW	16.856 mW
P_{total}	2.940 mW	17.653 mW

V. COMPARISON WITH REFERENCE

The FPGA results are directly comparable to [1] since both target the Zynq-7000 family. LUT count (1,221 vs. 980) is slightly higher due to the added run-time configurability and DSP count is nearly identical (25 vs. 26). The $2.6\times$ larger ASIC total area is explained by two factors: (i) the $130 \text{ nm} \rightarrow 65 \text{ nm}$ node ratio gives $\sim 4\times$ gate area scaling; (ii) the FF-based RAM adds $676,371 \mu\text{m}^2$ absent in the reference. Technology-scaling the logic-only area gives $178,473 \times (65/130)^2 \approx 44,618 \mu\text{m}^2$, consistent with the NTT datapath. The $5.8\times$ higher dynamic power is attributable to higher gate capacitance at 130 nm and the switching overhead of 33,500 FFs replacing a compact SRAM macro. Qualitative trends — NTT^{-1} drawing more power than NTT, n -independence of dynamic power, and the double Montgomery-multiply critical path — all match [1].

VI. CONCLUSION

A partial RTL-to-GDS implementation of [1] was realised on UMC 130 nm, achieving timing closure at 25 MHz with 17.65 mW post-route power. FPGA prototyping on the Zynq-7020 — the same device family as the reference — confirmed n -independent resource utilisation (1,221 LUTs, 25 DSPs) and correct $\text{NTT}/\text{NTT}^{-1}$ operation for all three PQC parameter sets, with LUT counts closely matching [1]. ASIC area and power differences are fully accounted for by technology-node scaling and the absence of an SRAM macro. Future work includes integrating an SRAM macro and targeting a more advanced process node. 70% of the reference paper[1] was implemented.

REFERENCES

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